Year 13 Maths - Projectiles

Similar questions - Projection from above ground level

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A golfer hits a ball from a point A, which is 5 m above the sea, with speed 20 $\mathrm{m\,s^{-1}}$ at 30° above the horizontal. Model the ball as a particle moving freely under gravity with $g = 10$. The ball lands in the sea at B.

(a) Show that the time taken to travel from A to B is $(1+\sqrt{2})$ seconds. **[4]**

(b) Find the horizontal distance of B from A. **[2]**

(c) Find the direction of motion of the ball 1.5 seconds after it is hit. **[3]**

(d) State one factor, other than air resistance, that could make the real distance different from your answer to (b). **[1]**

2 A stone is thrown horizontally at 12 $\mathrm{m\,s^{-1}}$ from the top of a vertical cliff 45 m high. Model the stone as a particle moving freely under gravity, $g = 10$.

(a) Find the time taken to reach the sea and the horizontal distance travelled. **[3]**

(b) Find the speed and direction of motion of the stone as it enters the sea. **[4]**

Question 1

(a) Vertical displacement $-5 = 20\sin 30^\circ t - 5t^2$, i.e. $-5 = 10t - 5t^2$, so $t^2 - 2t - 1 = 0$ and $t = 1 \pm \sqrt{2}$. Positive root $t = 1 + \sqrt{2}$

(b) $20\cos 30^\circ \times (1+\sqrt{2}) = 10\sqrt{3}(1+\sqrt{2}) = 41.8$ m

(c) $v_x = 10\sqrt{3} = 17.3$, $v_y = 10 - 15 = -5$. The ball is moving at $\arctan \dfrac{5}{17.3} = 16.1^\circ$ below the horizontal

(d) Any one: wind; the spin of the ball; the ball not being a particle (size, drag)

Question 2

(a) $45 = 5t^2$, $t = 3$ s. Distance $= 12 \times 3 = 36$ m

(b) $v_x = 12$, $v_y = 10 \times 3 = 30$. Speed $= \sqrt{144 + 900} = 32.3\ \mathrm{m\,s^{-1}}$ at $\arctan \dfrac{30}{12} = 68.2^\circ$ below the horizontal